Elias Hossain

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SUMMARY

Ph.D. researcher building LLM systems that stay reliable when human supervision does not scale. Every method runs **at inference time, on a model I cannot retrain**, because that is the access a deployed system actually has. A runtime monitor that screens an agent's plan before any tool executes and decides in 0.205 ms. A verification layer that cut a 7B model's refusal rate from 14.8% to 0.5% without moving win-rate. Channel gates that block tool-output poisoning outright and halve prompt-injection success. Each result ships with a public benchmark and a reference implementation, and is stress-tested in clinical settings where a confident error is measured in patient harm.

- **987 citations, h-index 13, i10-index 13** (Google Scholar, Sep 2026); **12+ first-author peer-reviewed papers**.
- First-author LLM safety work in **Findings of AACL-IJCNLP 2026**, at the **ICLR 2026 Trustworthy AI Workshop**, and in *IEEE T-SMC*; invited poster at the **Amazon Trusted AI Symposium**.
- **5 public benchmarks** (74,590+ records) and **14+ open-source repositories** on Hugging Face and GitHub.

EDUCATION

University of Central Florida, Orlando, FL — Aug 2025 – Present

Ph.D., Industrial Engineering & Management Systems, Computer Science concentration

- Advisor: Dr. Niloofar Yousefi; co-advisor: Dr. Ivan Garibay. Complex Adaptive Systems Laboratory.
- Dissertation: *Self-Verifying Scientific AI: A Verification-Centric Framework for Trustworthy Biomedical Reasoning*. Qualifying exam passed Summer 2026; expected graduation Spring 2028.

Mississippi State University, Mississippi State, MS — Jan 2024 – Jul 2025

M.S., Computer Science (Research), GPA 3.88/4.00. Advisors: Dr. Andy D. Perkins, Dr. Shahram Rahimi.

SELECTED PUBLICATIONS

First author unless noted. Full list on Google Scholar.

LLM Safety, Alignment, and Trustworthy AI

"NEXUS: Structured Runtime Safety for Tool-Using LLM Agents." *Findings of the Association for Computational Linguistics: AACL-IJCNLP 2026.* [arXiv, page] Plan-level monitor that screens an agent's structured plan before any tool executes and routes it to allow, block, confirm, or revise; +27.3 points intervention accuracy over rule-only monitoring at 0.205 ms median latency, under 0.1% overhead per agent turn.

"Fault-Tolerant Preference Alignment via Multi-Agent Verification." *ICLR 2026 Workshop on Principled Design for Trustworthy AI.* [OpenReview, page] Four heterogeneous verifier agents gate preference data under a k-of-n consensus rule; consensus provably suppresses corrupted supervision at an exponential rate, cutting summarization refusals from 0.148 to 0.005 at parity win-rate on Qwen2-7B.

"Safe and Scalable Collaboration in Multi-Agent LLM Systems: A Comprehensive Review." *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2026. [link]

"Learning Robust Representations for Malicious Content Detection via Contrastive Sampling and Uncertainty Estimation." *9th International Conference on Information and Computer Technologies (ICICT)*, 2026. [arXiv]

Applied ML and Biomedical AI

"Natural Language Processing in Electronic Health Records in relation to healthcare decision-making: A systematic review." *Computers in Biology and Medicine*, 2023. [link] **507 citations.**

"BIOGEN: evidence-grounded multi-agent reasoning framework for transcriptomic interpretation in antimicrobial resistance." *Frontiers in Bioinformatics*, 2026. [link, page]

"R-GAT: Graph Neural Networks for Cancer Document Classification." *Scientific Reports* (Nature Portfolio), 2026. [link]

S. Neupane, **E. Hossain**, et al. "From questions to insightful answers: Building an informed chatbot for university resources." *IEEE Frontiers in Education (FIE)*, 2024. [link] **63 citations.**

Additional: first-author review in *Briefings in Bioinformatics* (2026); multimodal brain-tumor segmentation, *Computers, Materials & Continua* (2022, **63 citations**); seven Scopus-indexed Springer Nature book chapters.

Manuscripts Under Review

First author on all.

"When Policies Cannot Be Retrained" (PoE-Steer). [arXiv] Closed-form, post-training steering of a frozen policy via product-of-experts composition, for deployments where fine-tuning is not an option. *NeurIPS 2026.*

"Right Knowledge, Wrong Answer: Test-Time Steering for Temporal Fact Conflicts in Open-Weight Language Models." [arXiv, page] Benchmark of 8,746 verified Wikidata transitions; activation patching localizes the outdated-fact preference to model-specific upper layers. *AAAI 2026.*

"Agreement Overstates Evidence: Error Dependence in LLM Judge Consensus." [page] Large-scale audit of whether LLM-judge banks fail independently, the assumption behind k-of-n consensus filtering. Supervised by Prof. Ser-Nam Lim (ex-Meta AI). *ICLR 2027.*

"ChannelGuard: Safe Models Do Not Compose into Safe Multi-Agent Systems." [arXiv, page] Training-free information-bottleneck gates on every inter-agent channel; per-trace attribution shows undefended pipelines outsource safety to provider filters. *KDD 2027.*

"UAT-LITE: Inference-Time Uncertainty-Aware Attention for Pretrained Transformers." [arXiv, page] *ACL ARR meta-review: borderline conference accept; committed to EACL 2027.*

Research Experience

Graduate Research Assistant, Complex Adaptive Systems Lab, UCF — Aug 2025 – Present

- Shipped **NEXUS**, a plan-level monitor that screens a tool-using agent's structured plan before any tool executes and routes it to allow, block, confirm, or revise: **+27.3 points** intervention accuracy over rule-only monitoring, 0.949 F_1, and a **0.205 ms** median decision (about 4,300 decisions/sec, under 0.1% overhead per agent turn). Accepted to Findings of AACL-IJCNLP 2026.
- Argued that safety properties of individual models do not compose, then built the defense: **ChannelGuard** places training-free information-bottleneck gates on every inter-agent channel. Across 2,100 traces and 8 attack families it **blocks tool-output poisoning outright and halves prompt-injection success** while holding task accuracy, on three model backends.
- Identified that RLHF and DPO train on preference data nobody checks, and fixed it upstream: four heterogeneous verifier agents under a k-of-n consensus rule cut Qwen2-7B's summarization refusal rate **from 14.8% to 0.5%** at parity win-rate, with a proof that consensus suppresses corrupted supervision exponentially.

- Released **5 public benchmarks (74,590+ records)** on Hugging Face, including CareBench (1,000 clinical agent trajectories across 5 frontier models) and a 9,250-item temporal-conflict QA set; ran multi-GPU training and evaluation across the Qwen, Llama, and Mistral open-weight families.

Graduate Research Assistant, Mississippi State University — May 2025 – Jul 2025

- Built a contrastive-learning detector for rare malicious events in large-scale security logs, pairing contrastive sampling with uncertainty estimation to hold precision stable under heavy class imbalance where standard training collapses to the majority class.
- Owned the retrieval and evaluation layer of BarkPlug V.2, a retrieval-augmented assistant **deployed campus-wide** at Mississippi State: FAISS retrieval, RAGAS-based answer scoring, and a System Usability Scale study with real students.

Industry Experience

Software Engineer, REVE Systems, Dhaka, Bangladesh — Dec 2022 – Nov 2023

Took speech recognition from research to production: cut word error rate **20%** and inference time **15%** on DeepSpeech and Wav2Vec 2.0, then shipped that stack as a voice interface for Bangladesh government e-governance services used by non-English-literate citizens.

AI/ML Researcher, Time Research & Innovation, UK — Oct 2020 – Nov 2022

Built the diagnostic models behind a COVID-19 screening pipeline over X-ray and CT imagery, and owned the machine-learning layer of a telemedicine platform delivered for UK healthcare.

Open-Source Software & Public Benchmarks

Benchmarks (huggingface.co/EliasHossain): nanobubbleeval (51,566 records, schema-constrained biomedical extraction); BioDivergence-Silver-v1.0 (11,900 claim pairs); ptc-benchmark (9,250 temporal-conflict QA); carebench (clinical agent trajectories); CancerAbstracts (1,874 abstracts).

Code (github.com/eliashossain001): reference implementations for MPV, ChannelGuard, NEXUS, UAT-LITE, TAS, and CareBench, each with a project page carrying method figures and result tables; also a 0.6B transformer trained from scratch in PyTorch and domain-adapted clinical language models.

Talks, Honors & Service

Talks. ICLR 2026 Workshop on Principled Design for Trustworthy AI (contributed talk); Amazon Trusted AI Symposium, JFK27 New York, 2026 (invited poster); Molecular Machine Learning Conference (MoML), MIT Jameel Clinic, 2025.

Honors. Graduate Presentation Fellowship, UCF, 2025 (selective award funding presentation at MoML, MIT Jameel Clinic). Finalist, Dr. Pradeep P. Thevannoor Innovation Awards, India, 2019.

Service. Reviewer, ICLR 2026 Trustworthy AI Workshop. Journal peer reviewer (2024–present) for seven venues including *Scientific Reports* (Nature Portfolio), *Computers in Biology and Medicine* (Elsevier), and *PLOS ONE*.

Teaching. Graduate Teaching Assistant, UCF (STA 3032: Probability and Statistics for Engineers), 2025–present; Mississippi State University, five graduate courses in software testing, computer forensics, machine learning, and AI, 2024–2025.

Technical Skills

LLMs & alignment: RLHF, DPO, SFT, LoRA/QLoRA; agentic and multi-agent systems; inference-time steering of frozen models; continual pretraining; multimodal modeling. **Trustworthy AI:** uncertainty quantification, calibration, Bayesian inference; LLM evaluation and benchmarking; runtime safety monitoring; adversarial robustness; graph neural networks. **Tooling:** Python, PyTorch, Hugging Face Transformers, scikit-learn, multi-GPU training, FAISS, RAG pipelines, SQL, R, Docker, Git, Azure, GCP.